

Designing a Software Solution Using Modern Development Techniques

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ASSIGNMENT

Title: Designing a Software Solution Using Modern Development Techniques

To do:

- Identify a real-world problem in your environment (e.g., managing hostel allocations, tracking attendance, or organizing student group projects).
- Propose a **software solution** to address the problem.
- Apply at least **three techniques in software development** (examples: modular programming, object-oriented design, version control, test-driven development, continuous integration).
- Show how each technique improves the quality, maintainability, or efficiency of the solution.

Deliverables:

- A design document including:
 - Problem statement
 - Proposed solution architecture (with UML diagrams or flowcharts)
 - Explanation of chosen techniques and their application
- A prototype or partial implementation (optional but encouraged).

SOLUTIONS

Problem Statement

I remember the last project that I did, it was in part 3 and we had an issue of assigning tasks to group members, we had to share the tasks unequally and that got some people upset. And since most of our communications were done online using whatsapp, there was lack of proper communications between the group members which caused us to almost miss the submission deadline. Therefore, there is a need for a software system that can help to share tasks equally and efficiently, which can also ensure there is proper documentation of tasks so as to track the progress of each group member.

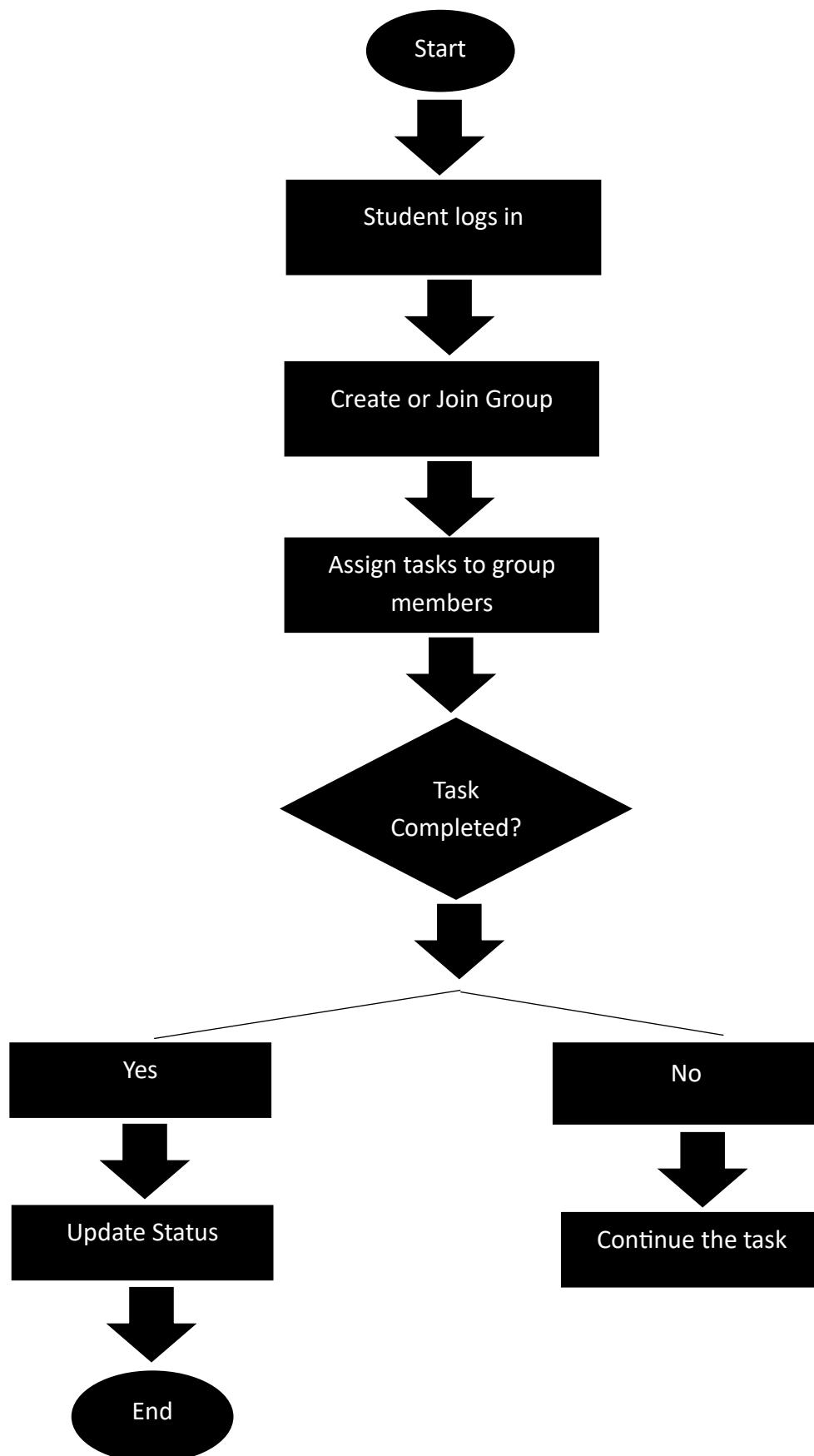
Proposed Software Architecture

To propose a software architecture, we first have to understand what it is we want to build.

The proposed software should be a software that can:

- Create groups for group members
- Share tasks efficiently
- Monitor each group member progress
- Keep track of the deadline for submission
- Mark as done once the task is completed

Flow Chart



Techniques used in the software design

1. Modularity Programming:

This is when the software involved is broken down into smaller independent modules which perform specific functions. How it is applied in our task management system is that each core function is assigned to a module which can work independently eg task management, authentication. This allows for faster development of the software and it makes it more efficient.

2. Object Oriented Programming:

Object oriented programming is a programming technique that involves organizing software using object and classes rather than functions. It is used to model real world entity effectively. Object oriented programming that makes large systems easy to manage and make code reusable and makes the code organized.

3. Version Control: It is used to help in tracking and managing changes in code. It is applied by making it easier for each group member to be able to track the progress of other members. It also makes it easier to collaborate with each group member.